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NUMERICAL & STATISTICAL METHODS

ASSIGNMENT

SUBMITTED BY,

NAME: MEENAKSHI PRAMOD

BATCH: CS-B (S5)

ROLL NO: 63 (20222067)

GAUSS'S FORWARD INTERPOLATION FORMULA

- Gauss's forward interpolation formula is,

$$F(x) = y_x = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!}\Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!}\Delta^4 y_{-2} + \cdots\cdots$$

PROOF

We know that, by the Newton's Gregory forward interpolation formula, we have

$$F(x) = y(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!}\Delta^4 y_0 + \cdots\cdots \quad -(1)$$

Now,

$$\Delta^2 y_0 = \Delta^2 E\, y_{-1} = \Delta^2(1+\Delta)y_{-1} = \Delta^2 y_{-1} + \Delta^3 y_{-1}$$

$$\Delta^3 y_0 = \Delta^3 E y_{-1} = \Delta^3(1+\Delta)y_{-1} = \Delta^3 y_{-1} + \Delta^4 y_{-1}$$

$$\Delta^4 y_0 = \Delta^4 E y_{-1} = \Delta^4(1+\Delta)y_{-1} = \Delta^5 y_{-1} + \Delta^4 y_{-1}$$

Similarly, $\Delta^4 y_{-1} = \Delta^4 y_{-2} + \Delta^5 y_{-2}$ and so on.

$$\Delta^n y_0 = \Delta^n y_{-1} + \Delta^{n+1} y_{-1}$$

Substituting the values of $\Delta^2 y_0$ and $\Delta^3 y_0$ in (1), we have,

$$F(x) = y(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}(\Delta^2 y_{-1} + \Delta^3 y_{-1}) + \frac{u(u-1)(u-2)}{3!}(\Delta^3 y_{-1} + \Delta^4 y_{-1}) + \frac{u(u-1)(u-2)(u-3)}{4!}(\Delta^4 y_{-1} + \Delta^5 y_{-1}) + \cdots$$

$$\Rightarrow F(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_{-1} + \left[\frac{u(u-1)}{2!} + \frac{u(u-1)(u-2)}{3!}\right]\Delta^3 y_{-1} + \left[\frac{u(u-1)(u-2)}{3!} + \frac{u(u-1)(u-2)(u-3)}{4!}\right]\Delta^4 y_{-1} + \cdots\cdots$$

$$\Rightarrow F(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_{-1} + \frac{u(u-1)}{3!}[3+u-2]\Delta^3 y_{-1} + \frac{u(u-1)(u-2)}{4!}[4+u-3]\Delta^4 y_{-1} + \cdots$$

$$\Rightarrow F(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!}\Delta^3 y_{-1} + \frac{u(u-1)(u-2)(u+1)}{4!}\Delta^4 y_{-1} + \cdots$$

Now,

$$\Delta^4 y_{-1} = \Delta^4 y_{-2} + \Delta^5 y_{-2}.$$

We, get the required result,

$$F(x) = y_x = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!}\Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!}\Delta^4 y_{-2} + \cdots$$

Hence, Gauss's forward interpolation formula proven.

NOTE:

1. This formula is known as Gauss's forward interpolation formula.
2. This formula involves odd differences below the central line ($x=a$) and even differences on the line.
3. Taking the central line and the next line, we have the differences occurring in the formula.

Central line: y_0 .... $\Delta^2 y_{-1}$ .... $\Delta^4 y_{-2}$ .... $\Delta^6 y_{-3}$

Next line: .... Δy_0 .... $\Delta^3 y_{-1}$ .... $\Delta^5 y_{-2}$

4. The formula is useful when u lies between 0 and 1.

GAUSS'S BACKWARD INTERPOLATION FORMULA

- The Gauss's Backward interpolation formula is,

$$F(x) = y(x_0+uh) = y_0 + \frac{u}{1!}\Delta y_{-1} + \frac{u(u+1)}{2!}\Delta^2 y_{-1} + \frac{(u-1)u(u+1)}{3!}\Delta^3 y_{-2}$$

$$+ \frac{(u+2)(u+1)u(u-1)}{4!}\Delta^4 y_{-2} + \cdots\cdots$$

where $u = \frac{x-x_0}{h}$.

PROOF

Starting with Newton-Gregory forward interpolation formula at $x = x_0$, we have,

$$y(x) = y(x_0+uh) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \cdots \quad (1)$$

Since, $\Delta y_0 = \Delta E y_{-1} = \Delta(1+\Delta)y_{-1} = \Delta y_{-1} + \Delta^2 y_{-1}$.

Similarly, $\Delta^2 y_0 = \Delta^2 y_{-1} + \Delta^3 y_{-1}$

$\Delta^3 y_0 = \Delta^3 y_{-1} + \Delta^4 y_{-1}$

$\Delta^3 y_{-1} = \Delta^3 y_{-2} + \Delta^4 y_{-2}$ etc.

Substituting the values of $\Delta y_0, \Delta^2 y_0, \Delta^3 y_0 \ldots$ in (1) we get,

$$y(x) = y_0(x_0+uh) = y_0 + u(\Delta y_{-1} + \Delta^2 y_{-1}) + \frac{u(u-1)}{2!}(\Delta^2 y_{-1} + \Delta^3 y_{-1})$$

$$+ \frac{u(u-1)(u-2)}{3!}[\Delta^3 y_{-1} + \Delta^4 y_{-1}] + \frac{u(u-1)(u-2)(u-3)}{4!}(\Delta^4 y_{-1} + \Delta^5 y_{-1}) + \cdots$$

$$\Rightarrow y(x) = y_0 + u\Delta y_{-1} + \left[u + \frac{u(u-1)}{2!}\right]\Delta^2 y_{-1} + \left[\frac{u(u-1)}{2!} + \frac{u(u-1)(u-2)}{3!}\right]\Delta^3 y_{-1}$$

$$+ \left[\frac{u(u-1)(u-2)}{3!} + \frac{u(u-1)(u-2)(u-3)}{4!}\right]\Delta^4 y_{-1} + \cdots$$

$$= y_0 + u\Delta y_{-1} + \frac{u}{2!}(2+u-1)\Delta^2 y_{-1} + \frac{u(u-1)}{3!}[3+u-2]\Delta^3 y_{-1}$$

$$+ \frac{u(u-1)(u-2)}{4!}[4+u-3]\Delta^4 y_{-1}$$

$$F(x) = y_0 + u\Delta y_{-1} + \frac{u(u+1)}{2!}\Delta^2 y_{-1} + \frac{u(u-1)(u+1)}{3!}\Delta^3 y_{-1} + \frac{u(u-1)(u-2)(u+1)}{4!}\Delta^4 y_{-1} + \cdots\cdots$$

Reduce the suffix by 1, except first three terms,

$$f(x) = y_0 + u\Delta y_{-1} + \frac{u(u+1)}{2!}\Delta^2 y_{-1} + \frac{(u-1)u(u+1)}{3!}\left[\Delta^3 y_{-2} + \Delta^4 y_{-2}\right] + \frac{(u-2)(u-1)u(u+1)}{4!}\left(\Delta^4 y_{-2} + \Delta^5 y_{-2}\right) + \cdots\cdots$$

$$f(x) = y_0 + u\Delta y_{-1} + \frac{u(u+1)}{2!}\Delta^2 y_{-1} + \frac{(u-1)u(u+1)}{3!}\Delta^3 y_{-2} + \frac{(u-1)u(u+1)}{4!}\left[4 + u - 2\right]\Delta^4 y_{-2} + \cdots\cdots$$

$$F(x) = y_0 + u\Delta y_{-1} + \frac{u(u+1)}{2!}\Delta^2 y_{-1} + \frac{(u-1)u(u+1)}{3!}\Delta^3 y_{-2} + \frac{(u-1)u(u+1)(u+2)}{4!}\Delta^4 y_{-2} + \cdots\cdots$$

Hence proved.

NOTE:

1. The above formula is known as Gauss's backward formula for interpolation.
2. This formula involves odd differences above the central line and even differences on the centra line.
3. Previous line: ..... Δy_{-1} ..... $\Delta^3 y_{-2}$ ..... $\Delta^5 y_{-3}$

   central line : ... y_0 ..... $\Delta^2 y_{-1}$ ..... $\Delta^4 y_{-2}$ ..... $\Delta^6 y_{-3}$
4. This formula is useful when u lies between -1 and 0.

1. Apply Gauss's forward central difference formula and estimate f(32) from the following table:

x:	25	30	35	40
$y=f(x)$:	0.2707	0.3027	0.3386	0.3794

ans: Let 30 be origin, $h=5$.

$$u=\frac{x-x_0}{h}=\frac{32-30}{5}=0.4$$

x	u	y	Δy	$\Delta^2 y$	$\Delta^3 y$
25	-1	0.2707			
			0.0320		
30	0	$\boxed{0.3027}$		$\boxed{0.0039}$	
			$\boxed{0.0359}$		$\boxed{0.0010}$
35	1	0.3386		0.0049	
			0.0408		
40	2	0.3794			

By Gauss's forward formula, we have,

$$y(x)=y(x_0+uh)=y_0+u\Delta y_0+\frac{u(u+1)}{2!}\Delta^2 y_{-1}+\frac{(u-1)u(u+1)}{3!}\Delta^3 y_{-1}+\cdots$$

$$y(x=32)=y(u=0.4)=0.3027+(0.4\times 0.0359)+\frac{(0.4\times -0.6)}{2}\times 0.0039$$

$$+\frac{(1.4\times 0.4\times -0.6)}{6}\times 0.0010$$

$$=0.3027+0.01436-0.000468-0.00006$$

$$y(x=32)=0.31653.$$

2. Using the following table, apply Gauss's forward formula to get f(3.75).

x:	2.5	3.0	3.5	4.0	4.5	5.0
$f(x)$:	24.145	22.043	20.225	18.644	17.262	16.047

ans: Let us take $x=3.5$ as the origin.

Here, $h=0.5$; $u=\frac{x-x_0}{h}=\frac{3.75-3.5}{0.5}=0.5$.

We form the difference table below.

x	u	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
2.5	-2	24.145					
			-2.102				
				0.284			
3.0	-1	22.043					
			-1.818				
					-0.047		
						$\boxed{-0.009}$	
3.5	0	$\boxed{20.225}$		$\boxed{0.237}$			
			$\boxed{-1.581}$		$\boxed{-0.038}$		$\boxed{-0.003}$
4.0	1	18.644		0.199		0.006	
			-1.382		-0.032		
4.5	2	17.262		0.167			
			-1.215				
5.0	3	16.047					

Gauss's forward interpolation formula is,

$$y(x) = y(x_0+uh) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!}\Delta^3 y_{-1}$$

$$+\frac{(u+1)u(u-1)(u-2)}{4!}\Delta^4 y_{-2} + \cdots$$

$$= 20.225 + (0.5\times -1.581) + \frac{(0.5\times -0.5\times 0.237)}{2} + \frac{(1.5\times 0.5\times -0.5\times -0.038)}{6}$$

$$+\frac{(1.5\times 0.5\times -0.5\times -1.5\times 0.009)}{24} + \cdots\cdots$$

$$= 20.225 - 0.7905 - 0.029625 + 0.002375 + 0.000210937 - 0.0000351562$$

$y(x=3.75) = \underline{19.4074258}$

3. Apply Gauss's forward formula to obtain $f(x)$ at $x=3.5$ from the table below.

x:	2	3	4	5
$f(x)$:	2.626	3.454	4.784	6.986

ans: Take $x=3$ as the origin; here $h=1$. $u = \frac{x-x_0}{h} = \frac{3.5-3}{1} = 0.5$.

x	u	$y=f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$
2	-1	2.626			
			0.828		
				$\boxed{0.502}$	
3	0	$\boxed{3.454}$			$\boxed{0.37}$
			$\boxed{1.330}$		
				0.872	
4	+1	4.784			
			2.202		
5	2	6.986			

Gauss's forward interpolation formula,

$$y(x)=y(x_0+uh)=y_0+u\Delta y_0+\frac{u(u-1)}{2!}\Delta^2 y_{-1}+\frac{(u+1)u(u-1)}{3!}\Delta^3 y_{-1}$$
$$+\frac{(u+1)(u)(u-1)(u-2)}{4!}\Delta^4 y_{-2}+\cdots\cdots$$

$$y(u=0.5)=3.454+(0.5\times 1.330)+\frac{1}{2}(0.5\times -0.5\times 0.502)+\frac{1}{6}(1.5\times 0.5\times -0.5\times 0.37)$$
$$=3.454+0.665-0.06275-0.023125$$

$$y(u=0.5)=\underline{\underline{4.033125}}$$

4. Using Gauss's backward interpolation formula, find the population for the year 1936 given that

Year	x:	1901	1911	1921	1931	1941	1951
Population in thousands	y:	12	15	20	27	39	52

ans: Since we require at $x=1936$, take 1941 as the origin.

$h=10$; $u=\frac{x-x_0}{h}=\frac{1936-1941}{10}=-0.5$

x	u	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
1901	-4	12					
			3				
1911	-3	15		2			
			5		0		
1921	-2	20		2		3	
			7		3		-10
1931	-1	27		5		-7	
			$\boxed{12}$		$\boxed{-4}$		
1941	0	$\boxed{39}$		$\boxed{1}$			
			13				
1951	1	52					

By Gauss's backward interpolation formula,

$$y(x)=y(x_0+uh)=y+u\Delta y_{-1}+\frac{u(u+1)}{2!}\Delta^2 y_{-1}+\frac{(u-1)u(u+1)}{3!}\Delta^3 y_{-2}+\cdots\cdots$$

$y(u=-0.5) = 39 + (0.5\times 12) + \frac{(-0.5\times 0.5\times -1.5)\times -4}{6} + \frac{(0.5\times -0.5)}{2}\times 1$

$y(u=-0.5) = 33 - \frac{1}{8} - \frac{1}{4} = \underline{\underline{32.625}}$ (thousands)

5. Find the value of $\cos 51^\circ 42'$ by using Gauss's backward interpolation formula from the table given below.

x:	50°	51°	52°	53°	54°
$y=\cos x$:	0.6428	0.6293	0.6157	0.6018	0.5878.

ans. Since we require interpolation at $x = 51^\circ 42'$, we take the origin at 52° (backward); here $h=1$.

$$u = \frac{x-52^\circ}{1^\circ} = \frac{51^\circ 42' - 52^\circ}{1^\circ} = \frac{-18'}{60'} = -0.3.$$

x	u	$y=\cos x$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
50°	-2	0.6428				
			-0.0135			
51°	-1	0.6293		-0.0001		
					-0.0002	
			-0.0136			
52°	0	0.6157		-0.0003		0.000
			-0.0139			
					0.0002	
53°	1	0.6018				
				-0.0001		
			-0.0140			
54°	2	0.5878				

By Gauss's backward formula,

$$y(x) = y_0 + u\Delta y_{-1} + \frac{u(u+1)}{2!}\Delta^2 y_{-1} + \frac{(u-1)u(u+1)}{3!}\Delta^3 y_{-2} + \ldots\ldots$$

$$y(u=-0.3) = 0.6157 + (-0.3\times -0.0136) + \frac{1}{2}(0.7\times -0.3\times -0.0003)$$
$$+ \frac{1}{6}(0.7\times -0.3\times -1.3\times -0.0002)$$

$y(u=-0.3) = 0.6157 + 0.00408 + 0.0000315 - 0.000009 = \underline{\underline{0.6198}}$

$\therefore \cos 51^\circ 42' = \underline{0.6198}$

6. If $\sqrt{12500} = 111.803399$, $\sqrt{12510} = 111.848111$, $\sqrt{12520} = 111.892805$, $\sqrt{12530} = 111.937483$, find $\sqrt{12516}$ by Gauss's backward formula.

ans: Since $\sqrt{12516}$ is wanted by backward formula, then take $x = 12520$ as the origin.

Here $h = 10$, $u = \frac{x - x_0}{h} = \frac{12516 - 12520}{10} = -0.4$.

Here $y = \sqrt{x}$.

x	u	$y=\sqrt{x}$	Δy	$\Delta^2 y$	$\Delta^3 y$
12500	−2	111.803399			
			0.044712		
12510	−1	111.848111		−0.000018	
			$\boxed{0.044694}$		$\boxed{-0.000002}$
12520	0	$\boxed{111.892805}$		$\boxed{-0.000016}$	
			0.044678		
12530	1	111.937483			

~~ans:~~ By Gauss's backward formula, we have

$$y(x) = y(x_0 + uh) = y_0 + u\Delta y_{-1} + \frac{u(u+1)}{2!}\Delta^2 y_{-1} + \frac{(u-1)u(u+1)}{3!}\Delta^3 y_{-2} + \cdots$$

$$y(u=-0.4) = 111.892805 + (-0.4 \times 0.044694) + \frac{1}{2}(0.6 \times -0.4 \times -0.000016) + \frac{1}{6}(0.6 \times -0.4 \times -1.4 \times -0.0000002)$$

$y(u=-0.4) = \underline{111.8749294}$

$\sqrt{12516} = \underline{111.8749294}$